PROJECT DOCUMENTATION

District-wise Missing-Person Data Analysis using Python

Author	Adeena Salam
Project Type	Python-based Data Analysis
Development Environment	Jupyter Notebook
Analysis Period	2021–2022
Dataset Size	1,841 records and 22 original columns

A descriptive analytical study of reported missing-person records, focusing on temporal, geographical, age-wise, and gender-wise patterns.

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1. Project Overview

This project uses Python-based data analysis to examine a district-wise missing-person dataset. The workflow includes data inspection, preprocessing, exploratory data analysis (EDA), statistical summaries, group-based comparisons, correlation analysis, and visualizations. The objective is to organize the available records and communicate descriptive patterns across years, states, districts, gender categories, and age groups.

The findings describe the records contained in the dataset. Aggregate reported counts should not be interpreted on their own as population-adjusted risk or as a direct measure of the underlying incidence of missing-person cases.

2. Project Objectives

- Inspect the dataset structure, data types, completeness, and duplicate records.
- Create derived indicators for total missing persons and demographic categories.
- Compare aggregate reported counts by year, state, district, gender, and age group.
- Use statistical summaries and visualizations to identify patterns and variation.
- Document key findings, limitations, and opportunities for further analysis.

3. Tools & Technologies

Tool	Purpose
Python	Programming and analytical workflow
Pandas	Data loading, cleaning, grouping, and reshaping
NumPy	Numerical operations
Matplotlib	Chart construction and formatting
Seaborn	Statistical visualization
Jupyter Notebook	Interactive development and documentation

4. Dataset Description

The dataset contains 1,841 rows and 22 original columns, including year, state, district, registration-circle information, and demographic counts for male, female, and transgender missing persons across child and adult categories. The analysis period represented in the current dataset is 2021–2022. The source name or publication details were not specified in the project information available for this documentation.

Data quality checks recorded no missing values and no duplicate rows. The original data types were considered appropriate for the analysis. Derived columns were created to summarize total missing persons by gender and age category, including detailed child-age group indicators.

5. Project Workflow

Dataset Inspection → Data Quality Checks → Derived Indicators → Exploratory Data Analysis → GroupBy & Pivot Analysis → Correlation Analysis → 10 Visualizations → Insights & Report

6. Analysis & Visualizations

Exploratory analysis was used to summarize the dataset, compare demographic categories, examine variation between geographic areas, and inspect relationships among derived indicators. Ten non-repetitive visualizations were prepared using a consistent black-and-red theme for a cohesive presentation.

- Year-wise total missing persons — line chart
- Top 10 states by aggregate total — horizontal bar chart
- Gender-wise distribution — pie chart
- Children versus adults — bar chart
- Distribution of total missing persons — histogram with KDE
- Gender-wise distribution across records — box plot
- Male versus female counts — scatter plot
- Correlation matrix — heatmap
- Top 10 districts by aggregate total — horizontal bar chart
- Year-wise gender comparison — grouped bar chart

7. Key Findings

- The aggregate derived total was 389,844 for 2021 and 441,964 for 2022, showing a higher recorded total in 2022.
- The gender-wise aggregate counts were 558,450 female, 272,952 male, and 406 transgender missing-person records.
- The age-wise aggregate counts were 160,765 children and 670,831 adults.
- State-level totals varied considerably. Maharashtra was identified as the state with the highest aggregate total in the analysis.
- District-level comparisons also showed variation, with a smaller number of districts contributing comparatively high aggregate counts.
- The total-missing-person indicator showed strong correlations with adult, female, male, and child indicators. Because the total is constructed from component counts, these strong correlations are partly expected mathematically.
- The distribution of record-level totals appeared right-skewed, indicating that many observations had lower counts while fewer observations had comparatively high counts.

Interpretation note

These are descriptive findings from the dataset as analysed. Higher counts may reflect population size, reporting practices, or other factors not included in the dataset. Correlation does not establish causation, and the very small transgender aggregate warrants cautious interpretation.

8. How to Use the Project

- Open the analysis notebook in Jupyter Notebook.
- Ensure Python and the required libraries—Pandas, NumPy, Matplotlib, and Seaborn—are installed.
- Place the dataset in the project folder and confirm the file path used by the notebook.
- Run the notebook cells sequentially, from data loading through final observations.
- Review the tables, visualizations, and written observations together.

9. Limitations

- The current analysis covers only the years available in the dataset (2021–2022).
- Counts are not adjusted for state or district population size.
- The dataset does not provide enough contextual information to explain why cases differ across places or demographic groups.
- Correlations involving the total indicator are partly influenced by the way the total is calculated from its components.
- Reported counts should not be treated as a complete measure of underlying incidence without additional context.

10. Future Scope

- Extend the analysis to more years to explore longer-term changes.
- Calculate population-adjusted rates when reliable population denominators are available.
- Add geographical maps and interactive filtering through a dashboard.
- Integrate relevant contextual data and explore carefully validated statistical or time-series methods.
- Automate report generation as new records become available.

11. Conclusion

This project demonstrates the practical use of Python for inspecting, processing, analyzing, and visualizing district-wise missing-person data. The analysis highlights differences across years, gender and age categories, states, and districts. The visualizations provide an accessible summary of the available records, while the limitations emphasize the importance of context, population-adjusted comparisons, and cautious interpretation. Future work can build on this foundation through expanded datasets, additional variables, and interactive reporting.

12. Acknowledgement

I would like to acknowledge the learning resources and technical tools that supported the completion of this project. Working through the analysis provided practical experience in data preprocessing, exploratory analysis, statistical summaries, and data visualization.

13. Author

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Project: District-wise Missing-Person Data Analysis using Python
Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, and Jupyter Notebook

END OF PROJECT

From Data to Insight